

# Multilevel Analysis

KU Leuven

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# Agenda I

This course consists of 12 sessions. In sessions 1-9, we will alternate between lectures and hands-on exercises in R. The last three sessions are dedicated to open questions and student presentations.

Sessions:

- 03/05, 17h-21h: Introduction to hierarchical data structures
- 03/06, 17h-21h: The linear two-level model with random intercepts
- 03/07, 09h-13h: The linear two-level model with random intercepts and random slopes
- 03/12, 17h-21h: Cross-classified, multiple membership and generalized mixed models

# Agenda II

## Sessions (continued):

- 03/13, 17h-21h: Analyzing panel data
- 03/14, 09h-13h: The within-between specification
- 03/19, 17h-21h: Contextual effects in multilevel models
- 03/20, 17h-21h: Multilevel models with more than two levels, the within-between model for pooledcross-sectional data
- 03/21, 09h-13h: Comparing and testing in multilevel modeling, estimation techniques, advanced topics
- 03/26, 17h-21h: Open session: Q&A, talking about course assignments
- 03/27, 17h-21h: Presentation of research proposals from students
- 03/28, 09h-13h: Presentation of research proposals from students

We are using R for this class.

- You should have RStudio installed on your computer.
- Download and extract the file `KUL_MultilevelAnalysis24.zip` from the Toledo course.
  - all data and code are included in this folder
  - run the file `RunFirstToPrepareRAndData.R` first, it should set up R for the hands-on exercises
  - this document is written in R Markdown and also included in the zip-file

## **Session 7: Contextual effects in multilevel models**

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# Agenda Session 7

Today:

- Reminder: What have we learned from panel data analysis?
- Defining contextual effects in multilevel modeling
- The within-between specification in cross-sectional multilevel models
  - Explained variation within and between level-2 clusters
  - Interaction effects and the within-between specification
  - Wearing the panel lens one more time
- Exercise 12: Estimating cross-sectional within-between models in R
- Assignment preparation: Checking out the ESS data base

## Reminder: What Have We Learned From Panel Data Analysis?

A two-level random intercept (RI) multilevel model:

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + \gamma_1 z_{1j} + u_j + e_{ij}$$

A random effects (RE) panel model:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \gamma_1 z_{1i} + u_i + e_{it}$$

- Statistically, these are the same models
  - difference in notation (*ij* vs *it*) and terminology
  - in multilevel models, units of analysis at level-1
  - in panel models, units of analysis at level-2

## RE and RI - What's the Difference

The difference between panel and multilevel modeling lies in the way we use the hierarchical data structures to answer (different) research questions

- In panel data analysis  $u_i$  is called **unobserved heterogeneity**
  - It captures differences between individuals and appears to be a problem
  - This is why FE models get rid of it and RE models are only recommended when  $u_i$  is uncorrelated with the variables in the model
  - The main interest then lies in variables that vary over time ( $x_{it}$ )
  - In sum, panel data is mostly used to aim for 'causal identification'



## RE and RI - What's the Difference

The difference between panel and multilevel modeling lies in the way we use the hierarchical data structures to answer (different) research questions

- With multilevel modeling, we are very often interested in countries and  $u_j$  is called the **random intercept**
  - The random intercept captures the differences between level-2 clusters (i.e. countries) and it suddenly appears to be 'very interesting'
  - Multilevel modelers typically are interested in explaining differences between level-2 clusters (i.e. countries). This is, explaining the variation in  $u_j$
  - Often we have a specific interest in context-level effects ( $z_j$ )
  - In sum, multilevel models are somewhat focused on the higher levels (i.e. country-level differences and effects)

## Remember the FE model

- The FE model gets rid of unobserved heterogeneity ( $u_i$ ) and thereby controls for any unobserved time-constant characteristics
- This is achieved by the so called FE transformation, where

$$(y_{it} - \bar{y}_i) = (\beta_0 - \beta_0) + \beta_1(x_{1it} - \bar{x}_{1i}) + \gamma_1(z_{1i} - \bar{z}_{1i}) + (u_i - \bar{u}_i) + (e_{it} - \bar{e}_i)$$

reduces to:

$$(y_{it} - \bar{y}_i) = \beta_1(x_{1it} - \bar{x}_{1i}) + (e_{it} - \bar{e}_i)$$

- effect of  $x$  is identified only from variation within units over time
- effects of variables that do not vary over time ( $z$ ), cannot be estimated

## Remember the RE Model

The RE model:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \gamma_1 z_{1i} + u_i + e_{it}$$

- The RE model identifies effects from variation within and between units  $i$ 
  - it can estimate effects of time-constant variables ( $z$ )
  - it must, however, assume that unobserved heterogeneity  $u_i$  is uncorrelated with the variables in the model
  - $\beta_1$  is a 'mix' of the within- and between-effect of  $x$  (keep that in mind for today's session)

## Combining FE and RE

- The within-between formulation provides within- and between-estimates of  $x$ 's effect in a RE model:

$$y_{it} = \beta_0 + \beta_1^{WE}(x_{1it} - \bar{x}_i) + \beta_1^{BE}\bar{x}_i + \gamma_1 z_{1i} + u_i + e_{it}$$

- The Mundlak formulation also provides within-estimates of the effect of  $x$ :

$$y_{it} = \beta_0 + \beta_1^{WE}x_{1it} + \beta_1^{BWD}\bar{x}_i + \gamma_1 z_{1i} + u_i + e_{it}$$

, where  $\beta_1^{WE}$  is the within estimate, equivalent to the FE estimator, and  $\beta_1^{BWD}$  provides an estimate of the difference between within- and between effects ( $\beta_1^{BE} = \beta_1^{WE} + \beta_1^{BWD}$ )

## Keep this in Mind!

- The Mundlak formulation is very important to understand today's session on contextual effects!
- Keep in mind:
  - $\beta_1^{WE}$  is the within-effect, which 'panel people' trust to be the better estimate of the true ('causal') effect of  $x$
  - this is because it is estimated under control of any differences between individuals
  - $\beta_1^{BWD}$  provides an estimate of the difference between within and between effects

# Context Effects in Multilevel Models

- Multilevel models are (most) often applied because there is an interest in context effects:

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + \gamma_1 z_{1j} + u_j + e_{ij}$$

- Examples:
  - does immigration inflow influence anti-immigrant attitudes?
  - does the unemployment rate influence life-satisfaction?
  - is pupil success dependent on the pupil-to-teacher-ratio?
  - is pupil success dependent on the share of pupils whose mother tongue is not the native language?

## Remember the Data Matrix

- We denoted higher- (or context-) level variables as  $z_j$ . By definition, a higher-level variable  $z_j$  must be constant within any given cluster  $j$ :

| Pupil_i | School_j | ExamScore_ij | PrevScore_ij | Gender_ij | SchoolType_j |
|---------|----------|--------------|--------------|-----------|--------------|
| 1       | 1        | 75           | 69           | M         | State        |
| 2       | 1        | 71           | 73           | M         | State        |
| 3       | 1        | 91           | 88           | F         | State        |
| 1       | 2        | 68           | 59           | F         | Private      |
| 2       | 2        | 37           | 37           | M         | Private      |
| 3       | 2        | 67           | 61           | M         | Private      |
| 1       | 3        | 82           | 83           | F         | State        |
| 2       | 3        | 85           | 81           | F         | State        |

- if  $z_j$  not constant within clusters  $j$ , there is something wrong with the data
- easy in the two-level model, but with more complex structures, it can be difficult to determine the level at which a variable is measured (more tomorrow)

## Context vs. Compositional Effects

- We already discussed that context effects are those effects that remain after compositional effects have been controlled
  - compositional effects refer to the higher-level variance explained by the distribution of level-1 variables
  - we revisit our example of unemployment and life satisfaction in a second
- Before, let us introduce the distinction between two types of higher-level variables:
  - **global**: Genuine macro-level variables, which occur only at the context-level and cannot be operationalized from lower-level variables (e.g. teaching style, welfare regime of a country, NATO membership of a country, public transportation quality in a district)
  - **aggregate**: Derived by aggregating a level-1 variable (e.g. students' mean SES, median income in a country, average rent per sqm in a district)



## Context vs. Compositional Effects

- When your context-level variables of interest is an aggregated characteristic, you must control for compositional effects
- Remember our example of unemployment and life satisfaction:
  - does the unemployment rate influence life satisfaction?
  - a context hypothesis implies that people in countries with high unemployment rates are less satisfied, independent of their own unemployment status
  - it also implies that the differences between countries cannot (fully) be explained by individuals' employment status
- A context effect is an effect above and beyond the individual-level effects

## Let's Revisit Our Example

Example: Does the unemployment rate influence life satisfaction?

- Assume that employed have a score of 10 (most happy) and unemployed have a score of 0 (most unhappy). There are no other effects whatsoever
- Data:
  - Country A: 100 employed, 0 unemployed – average score = 10
  - Country B: 90 employed, 10 unemployed – average score = 9
  - Country C: 80 employed, 20 unemployed – average score = 8
- Level-1 equation:

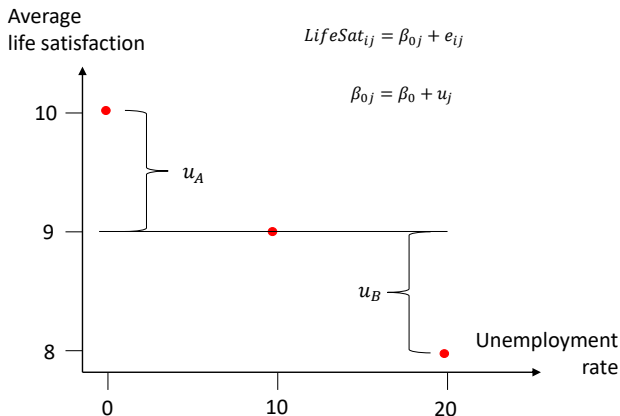
$$LifeSat_{ij} = \beta_{0j} + e_{ij}$$

- Level-2 equation:

$$\beta_{0j} = \beta_0 + \gamma_1 UnempRate_j + u_j$$

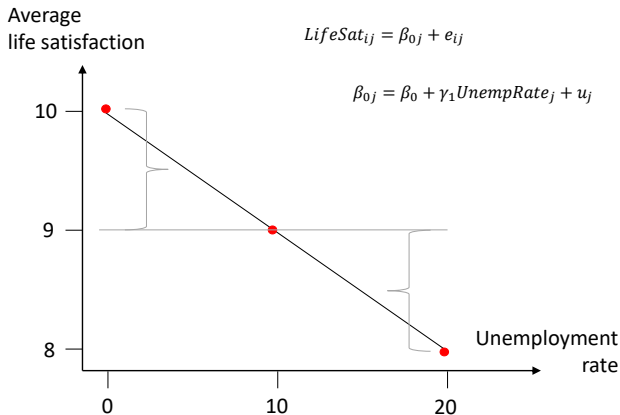
## Example: Unemployment and Life Satisfaction - Null Model

- The empty model would attribute all differences to countries (level-2)



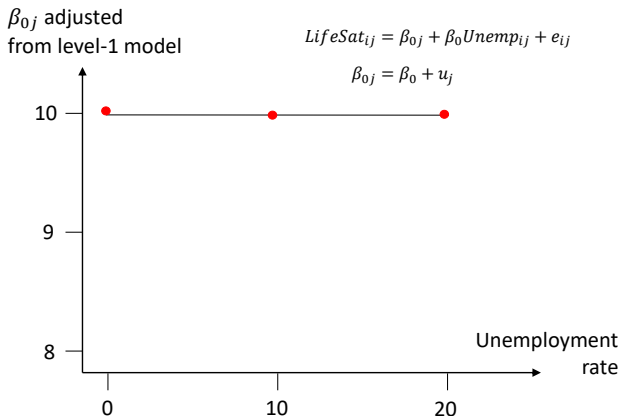
## Example: Adding Unemployment Rate

- A model with the unemployment rate as a level-2 variable would explain any differences between countries



## Example: Adding Individual Employment

- But so does a model with the individual unemployment rate. It leaves no room for a country-level effect of unemployment



## Example: Unemployment and Life Satisfaction

- Relationship between unemployment rate and average life satisfaction is due to an individual-level effect of unemployment
- The difference in average life satisfaction between the countries is merely due to a difference in the composition of their inhabitants, i.e. the individual-level distribution of unemployment
- Corollary: Whenever, you are interested in an aggregated higher-level variable, you must control for individual-level compositional effects of that characteristic and test whether there is an additional effect of the aggregated variable, i.e. a true context effect

## Let's Change Perspectives

- The unemployment examples was making the point that it is important to control for individual-level effects before introducing context effects
- But is it okay to identify individual-level effects without controlling for the context-level effects of aggregated variables?
- Revisiting the example:
  - As before, individual unemployment has an effect on life satisfaction (employed score 10, unemployed score 0)
  - Additionally, an increase of the unemployment rate by 10 percentage points, decreases life satisfaction by 1 point
  - What happens if we do not control for the unemployment rate at level-2? We will over-estimate the individual-level effect of unemployment!

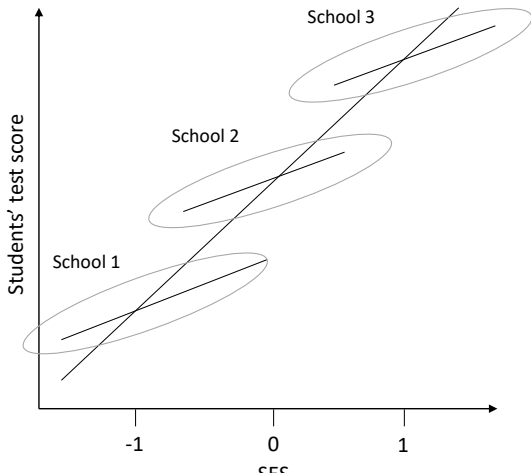
## Another Example: Students Attainment and SES

- A typical research question in educational research: Does student's socio-economic status (SES) influence their attainment?
  - how strong exactly is the effect?
  - how can the effect be reduced?
  - are there any context-level effects related to average SES in a class?



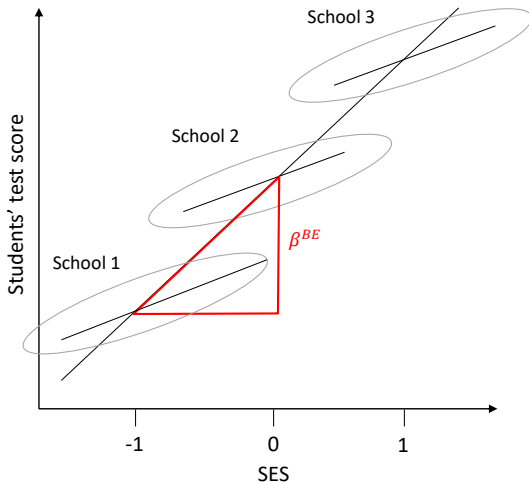
## Example: Student Attainment and SES

- what is the effect of SES?
- what does the context do here?
- what is the student-level effect of SES?



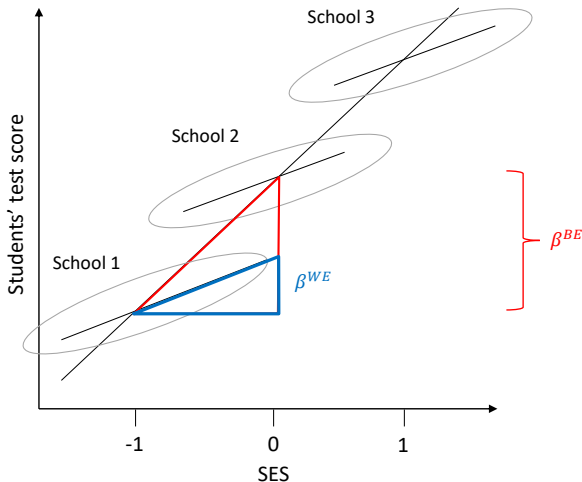
## Example: Student Attainment and SES - Between Effects

The between effect ( $\beta^{BE}$ ) is the difference in the mean score of two schools that differ by one in their mean SES



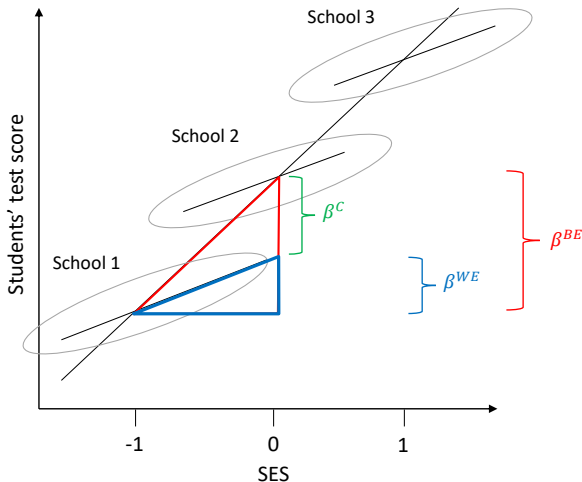
## Example: Student Attainment and SES - Within Effects

The within effect ( $\beta^{WE}$ ) is the difference between two students that differ by one in their SES score, but visit the **same** school



## Example: Student Attainment and SES - Context Effects

What's the context doing? It's the difference between two students with **same** SES in two schools which differ by one in mean SES score



## How to Separate Within, Between and Context Effects?

- The context effect is defined as  $\beta^C = \beta^{BE} - \beta^{WE}$
- If context variables are directly aggregated from the level-1 data (e.g. students' mean SES, mean absolute income in a country), we derive at the Mundlak formulation, which we know from panel data models:

$$y_{ij} = \beta_0 + \beta_1^{WE} x_{1ij} + \beta_1^C \bar{x}_j + u_j + e_{ij}$$

- note: With panel data we named the coefficient  $\beta^C$ , which now is the context effect,  $\beta^{DWB}$ , which was the difference of within- and between estimates
- looking through our 'panel lens', we can conclude that  $\beta_1^{WE}$  is the better estimate of the true ('causal') effect of  $x_1$  (e.g. SES)

# The Within-Between Specification

- Alternatively, we can use the within-between specification:

$$y_{ij} = \beta_0 + \beta_1^{WE}(x_{1ij} - \bar{x}_j) + \beta_1^{BE}\bar{x}_j + u_j + e_{ij}$$

- it will provide the context effect only indirectly as the difference between within and between estimates:  $\beta^C = \beta^{BE} - \beta^{WE}$

## Which Specification - Within-Between Or Mundlak

- The within-between and the Mundlak model are equivalent, i.e. they have the same likelihood
- Thus, there is not the correct model, it's a matter of taste
- With panel data often a preference for within-between, as it naturally separates time-varying from time-constant effects
- With cross-sectional multilevel data often a preference for the Mundlak model, as it provides direct estimates of the context effects
- The within-between model is technically more stable, as the demeaning reduces collinearity

## What About the Notion of Compositional Effects

- In this context, does it make sense to speak about compositional effects?
- We still can start with a model that includes only individual-level variables:  $y_{ij} = \beta_0 + \beta_1 x_{1ij} + u_j + e_{ij}$ 
  - we can judge how much variation between clusters  $j$  is explained by  $x_1$
  - but we now know that  $\beta_1 \neq \beta_1^{WE}$ , instead  $\beta_1 = (1 - \rho_x)\beta_1^{WE} + \rho_x\beta_1^{BE}$ , where  $\rho_x$  is the VPC of an empty regression with  $x$  as the dependent variable
  - substituting  $\beta_1^{BE}$  with  $(\beta_1^{WE} + \beta_1^C)$  and rearranging gives:

$$\beta_1 = \beta_1^{WE} + \rho_x\beta_1^C$$

- Thus, compositional effects include the context effects emerging from the individual-level variables



- With cross-level interactions we want to know, how the effect of a level-1 variable depends on the context-level characteristics
  - it makes sense to view the within-cluster effect as the effect moderated by the cluster-level characteristics
  - in line with recommendation from the literature that lower-level variables in cross-level interactions should be group-mean centered (Enders and Tofighi 2007)
- Interaction effects between level-1 and level-2 variables can be decomposed as well

# Random Slopes

- Random slopes can be estimated only for the within-country effect
- How to do that is obvious in the within-between model:

$$y_{ij} = \beta_0 + \beta_1^{WE}(x_{1ij} - \bar{x}_j) + \beta_1^{BE}\bar{x}_j + u_{0j} + u_{1j}(x_{1ij} - \bar{x}_j) + e_{ij}$$

but less obvious in the Mundlak model:

$$y_{ij} = \beta_0 + \beta_1^{WE}x_{1ij} + \beta_1^C\bar{x}_j + u_{0j} + u_{1j}(x_{1ij} - \bar{x}_j) + e_{ij}$$

## Cross-level Interactions in the Mundlak Model

- Decomposition of interaction between level-1 and level-2 variable in the Mundlak model:

$$y_{ij} = \beta_0 + \beta_1^{WE} x_{1ij} + \beta_1^C \bar{x}_j + \beta_3 z_{1j} + \beta_4^{WE} (x_{1ij} z_{1j}) + \beta_5^C (\bar{x}_j z_{1j}) + u_{0j} + u_{1j}(x_{1ij} - \bar{x}_j) + e_{ij}$$

- $\beta_5^C$  describes how the context effect of  $x$  depends on other characteristics of the context ( $z$ )

## Cross-level Interactions in the Within-between Specification

- Decomposition of interaction between level-1 and level-2 variable in the within-between model:

$$y_{ij} = \beta_0 + \beta_1^{WE}(x_{1ij} - \bar{x}_j) + \beta_1^{BE}\bar{x}_j + \beta_3 z_{1j} + \beta_4^{WE}((x_{1ij} - \bar{x}_j)z_{1j}) \\ + \beta_5^{BE}(\bar{x}_j z_{1j}) + u_{0j} + u_{1j}(x_{1ij} - \bar{x}_j) + e_{ij}$$

- But decomposition not necessary to get the within interaction from  $\beta_4^{WE}$
- $\beta_5^{BE}$  is the between effect of the interaction

# What About Context-Level Variables from Other Data Sources

- Aggregating individual-level data is one way to ‘create’ context-level variables
- When available, it is better to get data from other data sources
- Example: Unemployment and life satisfaction
  - we could estimate the unemployment rate from the survey data
  - likely no good estimate, as we assume a representative sample of the work force
  - we can also get the unemployment rate from the OECD, Eurostat, World Bank, etc.
  - this should reduce the measurement error in comparison to the first approach
- Still, what we have learned before applies here as well:
  - not including the unemployment rate can lead to an overestimation of the individual-level effect
  - but the total effect of unemployment cannot be decomposed into two components

# Looking Through the 'Panel Lens'

- Effects of level-1 variables
  - the within-effect ( $\beta^{WE}$ ) is holding the context constant, so it is probably a better estimate of the true effect of a variable
  - within estimation is a general identification strategy that works in many contexts. Examples:
    - comparing within individuals before and after an event
    - comparing children within families to hold social background constant
    - comparing identical twins to hold genetic influences constant
- Effect of level-2 variables
  - 'multilevel people' define  $\beta^C$  as the context effect
  - but 'panel people' conclude from the significance of  $\beta^{DWB}$  about the existence of unobserved variables
  - what to make out of that?

## Exercise 12: Estimating Context Effects in R

- Open the PDF “Exercise 12”; it will walk you through an analysis task with the SNS data.
- After the exercise, you can start to look for data for your assignment.

**Session 8: Multilevel models with  
more than two levels, the  
within-between specification for  
pooled cross-sectional data**

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Today:

- Multilevel models with three levels
  - The three-level random intercept model
  - Residuals in three-level models
  - The three-level random intercept and random slope model
- Pooled Cross-sectional Survey Data
  - How to model the hierarchical structure with pooled cross-sectional survey data?
  - A within-between decomposition of context effects
  - A societal growth curve model

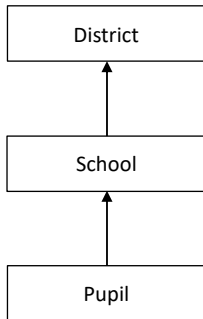
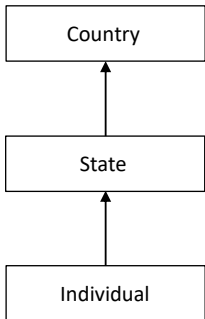
# Models We Have Discussed

- The two-level random intercept and random slope model
  - strict hierarchical nesting
  - units of analysis are nested in (one) higher-level cluster  $j$
- Cross-classified models with random intercepts
  - units of analysis are nested in multiple higher-level clusters ( $j$  and  $k$ )
  - but these are of different types
  - no cluster-levels above the higher-level clusters

# The Three-level Model

- Today, we discuss models where higher-level clusters are nested within additional clusters

Examples:



# The Three-level Random Intercept Model

- Remember the two-level random intercept model:

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + \gamma_1 z_{1j} + u_j + e_{ij}$$

- Expanding the model with an additional level above clusters  $j$ :

$$y_{ijk} = \beta_0 + \beta_1 x_{1ijk} + \gamma_1 z_{1jk} + \gamma_2 z_{2k} + v_{0k} + u_{0jk} + e_{ijk}$$

- $v_{0k}$  is the random intercept at level-3, i.e. of clusters  $k$
- $u_{0jk}$  is the random intercept at level-2, i.e. of clusters  $j$
- $z_{2k}$  is a variable describing level-3 clusters
- $z_{1jk}$  is a variable describing level-2 clusters

## Assumptions About the Error Term(s)

- Assumptions about the error term:
  - All error terms are IID (independent and identically distributed):

$$v_k \sim N(0, \sigma_{v0}^2), u_{jk} \sim N(0, \sigma_{u0}^2) \text{ and } e_{ijk} \sim N(0, \sigma_e^2)$$

- All error terms are independent of each other:

$$\text{Cov}(v_k, u_{jk}) = \text{Cov}(v_k, e_{ijk}) = \text{Cov}(u_{jk}, e_{ijk}) = 0$$

# Separate Equations

- Level-1 equation:

$$y_{ijk} = \beta_{0jk} + \beta_1 x_{1ijk} + e_{ijk}$$

- Level-2 equation:

$$\beta_{0jk} = \beta_{0k} + \gamma_1 z_{1jk} + u_{0jk}$$

- Level-3 equation:

$$\beta_{0k} = \beta_0 + \gamma_2 z_{2k} + v_{0k}$$

- the random intercept (RI) at level-2 varies around the RI at level-3
- thus, variation at level-2 ( $\sigma_{u0}^2$ ) is variation between level-2 clusters but within level-3 clusters

## Example: Students in Classes in Schools

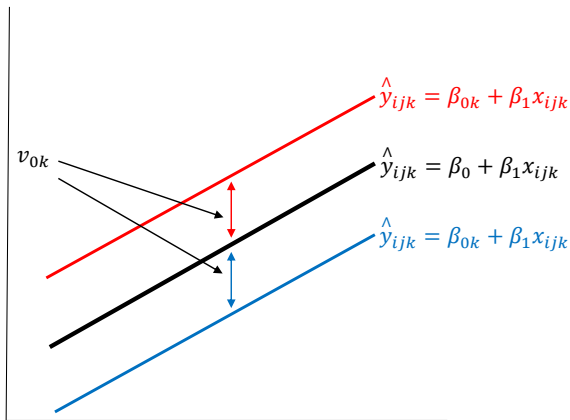
- Example: Students  $i$  are nested in classes  $j$  which are nested in schools  $k$
- An empty model:

$$y_{ijk} = \beta_0 + v_{0k} + u_{0jk} + e_{ijk}$$

- $v_{0k}$  is the school random effect, i.e. the schools deviation from the overall mean
- $u_{0jk}$  is the class random effect, i.e. the classes deviation from the school mean
- $e_{ijk}$  is the student-level residual, i.e. the students deviation from the class mean
- a student's deviation from the school mean, then, is  $u_{0jk} + e_{ijk}$

# Visualizing Residuals in Three-level Models

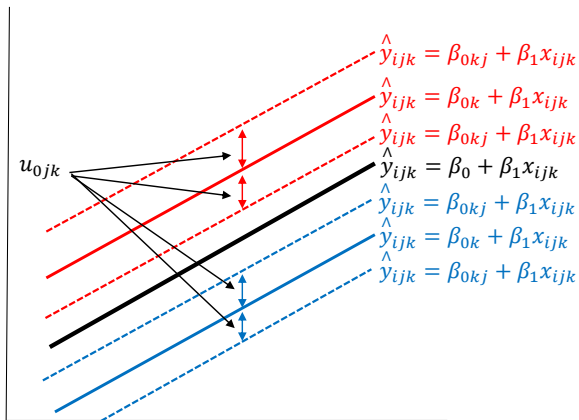
Residuals at level-3 (example: deviation of schools from average regression line)





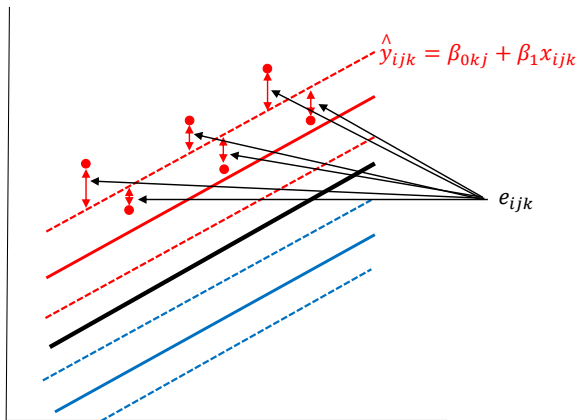
# Visualizing Residuals in Three-level Models

Residuals at level-2 (example: deviation of classes from school-specific regression line)



# Visualizing Residuals in Three-level Models

Residuals at level-3 (example: deviation of students from class-specific regression line)



# Variances in Three-level Models

- Variances in the three-level model
  - $\sigma_{v0}^2$  is the variance between level-3 clusters (e.g. schools)
  - $\sigma_{u0}^2$  is the variance between level-2 clusters (e.g. classes) and within level-3 clusters (e.g. schools)
  - $\sigma_e^2$  is the variance between level-1 units (e.g. students) and within level-2 clusters (e.g. classes) within level-3 clusters (e.g. schools)
  - $\sigma_{v0}^2 + \sigma_{u0}^2$  is the variance between level-2 units (e.g. classes)

# VPCs in the Three-level Model

- The VPC/ICC can be calculated in various ways
  - proportion of variance between schools (and intra-school correlation):

$$\frac{\sigma_{v0}^2}{\sigma_{v0}^2 + \sigma_{u0}^2 + \sigma_e^2}$$

- proportion of variance between classes (and intra-class correlation):

$$\frac{\sigma_{v0}^2 + \sigma_{u0}^2}{\sigma_{v0}^2 + \sigma_{u0}^2 + \sigma_e^2}$$

- proportion of variance between schools in relation to variance between classes:

$$\frac{\sigma_{v0}^2}{\sigma_{v0}^2 + \sigma_{u0}^2}$$

## R Example: Anti-immigrant Attitudes in Europe (2002-2018)

- We are using data from the ESS rounds 1 to 9
  - 22 countries
  - 9 survey rounds (every other year from 2002 to 2018)
- Such data has a three-level structure, where individuals are nested in country-years, which are nested in countries (we'll talk about this in detail later)
- Variables are identical to former examples (explaining anti-immigrant attitudes)
  - we use only left-right self-placement (a individual-level variable), and
  - unemployment rate (a country-year-level variable)

## R Example: Specifying Three-level Structures

- A strictly hierarchical three-level model is specified as a random interaction effect between level-3 and level-2 clusters, but leaving out the main effect of the level-2 clusters (compare this to the cross-classified model with random interaction)

```
# Estimate empty three-level model
model0 <- lmer(formula=ais ~ 1 + (1|Country) + (1|Country:essround),
               data=esslto9)

# Add individual-level variable
model1 <- lmer(formula=ais ~ 1 + lrscale + (1|Country) + (1|Country:essround),
               data=esslto9)

# Add country-year-level variable
model2 <- lmer(formula=ais ~ 1 + lrscale + unemp_rate_inter + (1|Country) + (1|Country:essround),
               data=esslto9)
```

**Table 2:** Three-level models of anti-immigrant sentiments

|                                   | M0                 | M1                    | M2                    |
|-----------------------------------|--------------------|-----------------------|-----------------------|
| (Intercept)                       | 0.0029<br>(0.0685) | 0.0021<br>(0.0688)    | 0.0014<br>(0.0680)    |
| lrscale                           |                    | 0.0971***<br>(0.0034) | 0.0971***<br>(0.0034) |
| unemp_rate_inter                  |                    |                       | 0.0311*<br>(0.0147)   |
| AIC                               | 224468.6594        | 223655.2529           | 223659.3843           |
| BIC                               | 224505.8981        | 223701.8013           | 223715.2424           |
| Log Likelihood                    | -112230.3297       | -111822.6264          | -111823.6921          |
| Num. obs.                         | 81608              | 81608                 | 81608                 |
| Num. groups: Country:essround     | 172                | 172                   | 172                   |
| Num. groups: Country              | 22                 | 22                    | 22                    |
| Var: Country:essround (Intercept) | 0.0224             | 0.0225                | 0.0222                |
| Var: Country (Intercept)          | 0.1000             | 0.1008                | 0.0983                |
| Var: Residual                     | 0.9106             | 0.9014                | 0.9014                |

\*\*\*  $p < 0.001$ ; \*\*  $p < 0.01$ ; \*  $p < 0.05$

## R Example: Variances

### Calculating variation between countries and between country-years

```
# Calculate proportion of variance between countries  
.1000/ (.1000+.0224+.9106)
```

```
## [1] 0.09680542
```

```
# Calculate proportion of variance between country-years  
.1000+.0224/ (.1000+.0224+.9106)
```

```
## [1] 0.1216844
```

```
# How much of higher-level variation is between countries  
.1000/ (.1000+.0224)
```

```
## [1] 0.8169935
```

- about 10% of total variation is between countries (level-3) and about 12% above level-1
  - 82% of higher-level variation is between countries
  - thus, not much change within countries over time



# The Random Slope Model

Extending the three-level model with random slopes:

$$y_{ijk} = \beta_0 + \beta_1 x_{1ijk} + \gamma_1 z_{1jk} + \gamma_2 z_{2k} + \\ v_{0k} + v_{1k} x_{1ijk} + v_{2k} z_{1jk} + u_{0jk} + u_{1jk} x_{1ijk} + e_{ijk}$$

## Random Slopes in Three-level Models

- Random slopes can be included for lower-level variables, not just for level-1 variables
  - $v_{2k}$  is the random slope of the variable  $z_{1jk}$
  - it is the variation in the effect of a level-2 variable ( $z_{1jk}$ ) between level-3 clusters ( $k$ )
  - example: varying effect of student-to-teacher ratio between schools
- For level-1 variables ( $x_{1ijk}$ ), random slopes can (should) be included at all higher-levels
  - $v_{1k}$  is the level-3 deviation from the overall slope ( $\beta_1$ )
  - $u_{1jk}$  is the level-2 deviation from the level-3 random effect
  - effect of  $x_{1ijk}$  in a given level-2 cluster ( $jk$ ) is  $\beta_1 + v_{1k} + u_{1jk}$

# R Example: Anti-immigrant Attitudes in Europe

*# Adding a random slope for left-right position*

```
model3 <- lmer(formula=ais ~ 1 + lrscale + unemp_rate_inter + (1 + lrscale|Country) +  
               (1 + lrscale|Country:essround), data=essito9)  
summary(model3)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: ais ~ 1 + lrscale + unemp_rate_inter + (1 + lrscale | Country) +  
##      (1 + lrscale | Country:essround)  
##      Data: essito9  
##  
## REML criterion at convergence: 221789.4  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max   
## -3.6379 -0.6721 -0.0654  0.5988  3.7706   
##  
## Random effects:  
##      Groups          Name          Variance Std.Dev. Corr   
## Country:essround (Intercept) 0.021131 0.14536   
##                  lrscale      0.003172 0.05632 -0.03   
## Country          (Intercept) 0.099572 0.31555   
##                  lrscale      0.020819 0.14429 -0.19   
## Residual                        0.878606 0.93734   
## Number of obs: 81608, groups: Country:essround, 172; Country, 22  
##  
## Fixed effects:  
##  
##              Estimate Std. Error      df t value Pr(>|t|)   
## (Intercept)    0.01153    0.06835  20.74216   0.169  0.86762   
## lrscale         0.09893    0.03130  20.94688   3.160  0.00473 **
```

## R Example: Looking at Results

**Table 3:** Three-level model with random slopes of left-right

|                                                | M3                   |
|------------------------------------------------|----------------------|
| (Intercept)                                    | 0.0115<br>(0.0683)   |
| lrscale                                        | 0.0989**<br>(0.0313) |
| unemp_rate_inter                               | 0.0307*<br>(0.0144)  |
| AIC                                            | 221809.3682          |
| BIC                                            | 221902.4650          |
| Log Likelihood                                 | -110894.6841         |
| Num. obs.                                      | 81608                |
| Num. groups: Country:essround                  | 172                  |
| Num. groups: Country                           | 22                   |
| Var: Country:essround (Intercept)              | 0.0211               |
| Var: Country:essround lrscale                  | 0.0032               |
| Cov: Country:essround (Intercept) lrscale      | -0.0002              |
| Var: Country (Intercept)                       | 0.0996               |
| Var: Country lrscale                           | 0.0208               |
| Cov: Country (Intercept) lrscale               | -0.0084              |
| Var: Residual                                  | 0.8786               |
| *** $p < 0.001$ ; ** $p < 0.01$ ; * $p < 0.05$ |                      |

## R Example: Interpretation

- Correlation between intercept and slope is negative, but only at the country level
  - 'fanning in' pattern exists between countries
  - within countries there is no pattern
- Effect of left-right varies substantially
  - most variation is between countries, where the standard deviation of the left-right slope is .14
  - much less variation within countries, where the standard deviation is .06
  - effect of left-right is mostly determined by the country-context and relatively stable across time

# Pooled Cross-sectional Survey Data

- Many international surveys have now been fielded multiple times (e.g. European Social Survey, International Social Survey Programme, World Values Survey)
  - everything on the following slides also applies to national cross-sectional survey data, where the context-level may be neighborhoods, regions or states
- It is common to pool the data from multiple waves. This gives a data set with three levels:
  - respondents are located at level 1
  - they are nested in countries
  - and they are nested in survey waves
- What is the correct model for such data?

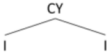
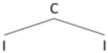
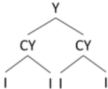
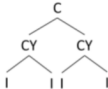
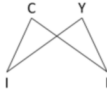
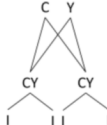
# How to Model Pooled Cross-sectional Survey Data?

Schmidt-Catran and Fairbrother (2016) discuss various modeling approaches:

*European Sociological Review*, 2016, Vol. 32, No. 1

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**Table 1.** A typology of random effects structures for multilevel models of comparative longitudinal survey data

| Random effects | Model A                                                                           | Model B                                                                           | Model C                                                                           | Model D                                                                           | Model E                                                                            | Model F                                                                             |
|----------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Country        |                                                                                   | ✓                                                                                 |                                                                                   | ✓                                                                                 | ✓                                                                                  | ✓                                                                                   |
| Year           |                                                                                   |                                                                                   | ✓                                                                                 |                                                                                   | ✓                                                                                  | ✓                                                                                   |
| Country-year   | ✓                                                                                 |                                                                                   | ✓                                                                                 | ✓                                                                                 |                                                                                    | ✓                                                                                   |
| Structure      |  |  |  |  |  |  |

Note: C=country-level RE, Y = year-level RE, CY = country-year-level RE, I = individual level.

# How to Not Model Pooled Cross-sectional Survey Data?

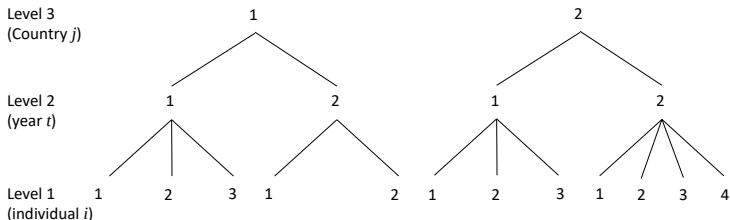
Schmidt-Catran and Fairbrother (2016) discuss various modeling approaches:

| ID  | FE      |      |              | RE structure |      |              |      | Author(s)                     | Year |
|-----|---------|------|--------------|--------------|------|--------------|------|-------------------------------|------|
|     | Country | Year | Country-Year | Country      | Year | Country-Year | Type |                               |      |
| 76  | ✓       |      | ✓            |              |      | ✓            | A    | Gerlitz <i>et al.</i>         | 2012 |
| 86  | ✓       |      | ✓            |              |      | ✓            | A    | van der Lippe <i>et al.</i>   | 2011 |
| 116 | ✓       |      | ✓            |              |      | ✓            | A    | Huijts <i>et al.</i>          | 2010 |
| 30  | ✓       |      |              | ✓            |      |              | B    | Immerzeel and van Tubergen    | 2013 |
| 44  | ✓       |      |              | ✓            |      |              | B    | Dinesen                       | 2013 |
| 62  | ✓       |      |              | ✓            |      |              | B    | Stegmueller <i>et al.</i>     | 2012 |
| 67  | ✓       |      | ✓            | ✓            |      |              | B    | Meulemann                     | 2012 |
| 78  | ✓       |      |              | ✓            |      |              | B    | Engelhardt                    | 2012 |
| 114 | ✓       |      |              | ✓            |      |              | B    | Kalmijn                       | 2010 |
| 118 |         |      | ✓            | ✓            |      |              | B    | Eger                          | 2010 |
| 149 | ✓       |      |              | ✓            |      |              | B    | Biedinger, Becker and Rohling | 2008 |
| 158 |         |      | ✓            | ✓            |      |              | B    | Fekjær and Birkelund          | 2007 |
| 166 |         |      | ✓            | ✓            |      |              | B    | Kogan and Kalter              | 2006 |
| 176 | ✓       | ✓    | ✓            | ✓            |      |              | B    | Meulemann                     | 2004 |
| 17  |         | ✓    | ✓            |              | ✓    | ✓            | C    | Schlueter and Davidov         | 2013 |
| 173 |         |      | ✓            |              | ✓    | ✓            | C    | Andersen <i>et al.</i>        | 2006 |
| 40  | ✓       |      | ✓            | ✓            |      | ✓            | D    | Fairbrother                   | 2013 |



# Pooled Cross-sectional Survey Data As a Three-level Structure

- They conclude that such data is most often best modeled with respondents, nested in years/waves, nested in countries (Type D):



# The Potential of Pooled Cross-Sectional Survey Data

- This structure has a nice property:
  - the levels 2 and 3 constitute a 'panel structure' with countries as the units of analysis, which are repeatedly observed
  - thus, we can analyse this data similar to pooled TSCD, using panel data models
  - at the same time, however, we can control for compositional effects from the individual level
  - and we can include cross-level interactions

## A Model for Pooled Cross-sectional Survey Data

A three-level model with individuals ( $i$ ), nested in time ( $t$ ), nested in countries ( $j$ )

$$y_{jti} = \beta_0 + \beta_1 x_{1jti} + \gamma_1 z_{1jt} + \gamma_2 z_{2j} + v_{0j} + u_{0jt} + e_{jti}$$

- Note: To indicate the panel-like structure, notation uses different order of indices (from highest to lowest level)
- $v_{0j}$  is the random intercept at level-3 (i.e. country-level) and  $u_{0jt}$  the random intercept at level-2 (i.e. country-year)
- The cluster-level variables (e.g. unemployment rate) can be divided into those that vary over time ( $z_{1jt}$ ) and those that are constant ( $z_{2j}$ )

## Decomposing the Cluster-level Effects

- As levels 2 and 3 have a panel structure, it is easy to get a within estimator of the country-level effects:

$$y_{jti} = \beta_0 + \beta_1 x_{1jti} + \gamma_1^{WE} (z_{1jt} - \bar{z}_{1j}) + \gamma_1^{BE} \bar{z}_{1j} + \gamma_2 z_{2j} + v_{0j} + u_{0jt} + e_{jti}$$

- This model decomposes the cluster-level effects ( $z$ ) into their within and between components
  - this is an important advantage over a cross-sectional multilevel model
  - we combine the virtues of panel data analysis, i.e. within-estimation, with the advantage of multilevel modeling, i.e. accounting for compositional effects
  - proposed by Fairbrother (2014), also discussed in Schmidt-Catran, Fairbrother, and Andreß (2019)
  - later extended for cross-level interactions by Giesselmann and Schmidt-Catran (2019)

## R Example: Decomposing Country-level Effects

```
# Transform variables for within-between model
ess1to9 <- ess1to9 %>% group_by(Country) %>%
mutate(
  unemp_m = mean(unemp_rate_inter, na.rm = TRUE),
) %>% ungroup() %>%
mutate(
  unemp_dm = unemp_rate_inter-unemp_m,
)

model3 <- lmer(formula=ais ~ 1 + lrscale + unemp_m + unemp_dm
               + (1|Country) + (1|Country:essround), data=ess1to9)
```

**Table 4:** Three-level model of anti-immigrant sentiments

|                                   | cross-sectional     | within-between      |
|-----------------------------------|---------------------|---------------------|
| (Intercept)                       | 0.001<br>(0.068)    | −0.000<br>(0.069)   |
| lrscale                           | 0.097***<br>(0.003) | 0.097***<br>(0.003) |
| unemp_rate_inter                  | 0.031*<br>(0.015)   |                     |
| unemp_m                           |                     | 0.084<br>(0.092)    |
| unemp_dm                          |                     | 0.030*<br>(0.015)   |
| AIC                               | 223659.384          | 223663.964          |
| BIC                               | 223715.242          | 223729.132          |
| Log Likelihood                    | −111823.692         | −111824.982         |
| Num. obs.                         | 81608               | 81608               |
| Num. groups: Country:essround     | 172                 | 172                 |
| Num. groups: Country              | 22                  | 22                  |
| Var: Country:essround (Intercept) | 0.022               | 0.022               |
| Var: Country (Intercept)          | 0.098               | 0.102               |
| Var: Residual                     | 0.901               | 0.901               |

\*\*\*  $p < 0.001$ ; \*\*  $p < 0.01$ ; \*  $p < 0.05$

## R Example: Interpretation

- Unemployment rate has a significant positive effect in a standard three-level model
- When decomposing the effect into between and within-country effects, the within country effect turns out to be significant, while the between country-effect is much larger but not significant

## Exercise 13: Three-level Models in R

- Open the PDF “Exercise 13”; it will walk you through two analysis tasks.



## **Session 9: Estimation techniques, comparing and testing in multilevel modeling, some advanced topics**

---

Today:

- How multilevel models are estimated
  - Bias in the random part
  - Full Maximum Likelihood vs. Restricted Maximum Likelihood
- Statistical testing and why it is difficult in multilevel models
- Testing single coefficients in the fixed part
- Comparing nested models using LR-Tests
  - Comparing models that differ in the fixed part
  - Comparing models that differ in the random part
- Regression diagnostics (very brief)

## What We Ignored Until Now

- We have never talked about parameter estimation
  - we just took the estimates, using the standard estimation algorithm of lmer (and glmer)
  - parameters in multilevel models (typically) do not have an analytical solution
  - they require numerical optimization in an iterative procedure
- Testing hypotheses and selecting the best model specification requires statistical tests
  - testing in multilevel models is a complex (and still debated) issue

# Estimation of Parameters in Multilevel Models

- Maximum Likelihood (ML) is a general estimation techniques, which can be applied to many models
- ML finds the parameters that make the observed data 'most likely'. More precisely, it finds the parameters that maximize the likelihood that the process described by the model produced the observed data
- ML is an iterative process, aiming to find the maximum of a (complex) function (the Likelihood function)
- In multilevel models, one iteration consists of two steps:
  - estimate parameters of fixed part
  - then, estimate random part, conditional on the updated fixed part

# Full Maximum Likelihood

- Asymptotically (if  $n \rightarrow \infty$ ), Full Maximum Likelihood (FML) is unbiased
- FML estimates, as described on the slide before, the random and the fixed part of the model simultaneously
  - this is, FML uses estimates from the fixed part (i.e. estimated means) to estimate the variances in the random part
  - it substitutes unknown parameters with an estimated value but does not correct for the degrees of freedom used for the estimation of the means
- Variance estimates in FML can be biased
  - specifically, FML tends to **underestimate** the variance components
  - the smaller  $n$  and the higher the dimensionality of the data, aka the number of variables, the stronger is the bias

## Restricted Maximum Likelihood

- Restricted Maximum Likelihood (REML) estimates the random and the fixed part separately
  - the random part is estimated first, using a linear transformation of the data, that makes the identification of variance parameters possible, without the use of estimated means
  - the fixed part is estimated, after the random effects are identified
- REML accounts for sampling variation (i.e. the fact that we do not know the true means) by a correction of the degrees of freedom
- REML has been specifically designed to avoid the bias from FML estimation (Patterson and Thompson 1971)
  - Schmidt-Catran, Fairbrother, and Andreß (2019) discuss FML and REML and point to the more detailed literature

- The estimation technique (FML or REML) is to be distinguished from the algorithm used to 'find' the parameters
- Many different optimizers, aka algorithms to find the maximum of the Likelihood function, are available
- Default optimizer in most packages are chosen well, but sometime it can be necessary to switch between optimizers
  - with very large data and complex models, time may be an issue. Some algorithms are faster than others
  - sometimes one optimizer does not converge, while an other does
  - with lmer the argument `lmerControl` allows to chose the optimizer
  - we do not go into the details here

## REML or FML?

- Sample size in multilevel models often an issue at the higher level(s)
- When  $N_j$  is small, FML will be biased and REML will be unbiased
- What is small?
  - hard to say as it also depends on the model complexity
  - bias of FML stronger in non-linear models than in the linear mixed model
  - Rule of thumb (linear case): If  $N_j > 20$ ; bias in FML is relatively small; if  $N_j < 15$ , it is severe (Elff et al. 2016)
  - Better to try both estimators, instead of relying on a simple rule of thumb



## Why Not Always REML?

- If FML can be biased, but REML is unbiased, why not always use REML?
- FML has one clear advantage over REML:
  - with FML we can use Likelihood-Ratio tests (LR tests) to compare nested models that differ in the fixed and/or random part of the model
  - with REML, where fixed and random part are estimated separately, we can use LR tests only on nested models which differ in their random part
- This suggests a two-fold strategy
  - use FML to build up and test the fixed part of your model
  - use REML to build up and test the random part of your model and to get final estimates
  - if bias in FML is small, you may just go with it

## R Example: Comparing FML and REML

We are using the ESS data from round 8 and explain anti-immigrant attitudes

```
# Estimate empty model with FML (note: the default of lmer is REML)  
model0_fml <- lmer(formula=ais ~ 1 + (1|Country),  
                   REML=FALSE, data=essround8)  
# Estimate empty model with REML  
# (note: this is the default, so REML=TRUE is not necessary)  
model0_reml <- lmer(formula=ais ~ 1 + (1|Country),  
                   REML=TRUE, data=essround8)
```

## R Example: Comparing Estimates from FML and REML

**Table 5:** Empty models of Anti-immigrant sentiments (FML vs. REML)

|                                                | FML                 | REML                |
|------------------------------------------------|---------------------|---------------------|
| (Intercept)                                    | −0.1236<br>(0.0947) | −0.1236<br>(0.0971) |
| AIC                                            | 26156.6531          | 26159.5047          |
| BIC                                            | 26178.0838          | 26180.9354          |
| Log Likelihood                                 | −13075.3265         | −13076.7523         |
| Num. obs.                                      | 9354                | 9354                |
| Num. groups: Country                           | 21                  | 21                  |
| Var: Country (Intercept)                       | 0.1860              | 0.1955              |
| Var: Residual                                  | 0.9492              | 0.9492              |
| *** $p < 0.001$ ; ** $p < 0.01$ ; * $p < 0.05$ |                     |                     |

- ICC with FML  $\frac{.1860}{.1860+.9492} = .1638$
- With REML  $\frac{.1955}{.1955+.9492} = .1707$

# What Is Statistical Testing?

- The null hypothesis  $H_0$  always assumes no difference (i.e. no effects)
- The alternative hypothesis  $H_1$  assumes difference (i.e. effects)
- Procedure of a statistical test:
  1. calculate a test statistic (depends on the nature of the problem)
  2. compare the test statistic with a (tabulated) distribution of a statistic under the assumption that  $H_0$  is true
  3. if the test statistic is—with some probability—greater than the statistic under the assumption of  $H_0$ , there is a significant differences (i.e. effect)
- Typically, we speak of significant effects if the probability to observe the test statistic under  $H_0$  is smaller than .05

# Statistical Testing in Regression Models

- There are three forms of testing in regression models
  - testing the overall model: Comparing a complex model with a simpler one; where the simplest model is the empty model
  - testing individual coefficients
  - testing groups of coefficients
- In standard OLS regression . . .
  - F-Test for linear restrictions to test the overall model (all coefficients—except intercept—are zero) and to test groups of coefficients
  - t-Test to test single coefficients
  - both, F-Test and t-test, are theoretically well understood, i.e. the model parameters ( $R^2$  and any coefficient  $\beta$ ) have (nice) asymptotic distributions against which we can test

## Testing in Multilevel Models

- Models with random effects, aka mixed or multilevel models, do not have these nice asymptotic distributions
- How many degrees of freedom does a multilevel model have?
  - Typically, one DF per estimated parameter
  - The design matrix of the REs, contains one indicator variable for every higher-level (e.g. 21 countries), but the model summarizes these indicators with one parameter (the variance)
  - So, how many degrees of freedom do we use for the random part (e.g. 1 or 21)
  - Shrinkage reduces the 'information used from each cluster  $j$ ' (e.g. so it is not 21, but what is it?)
  - Still debated how to calculate the degrees of freedom in multilevel models

## Testing Single (Fixed) Coefficients with lme4

- Because of the aforementioned difficulties, the authors of lme4 decided to not provide p-values for fixed effects estimates from lmer
  - they argue that the distributions of the test statistics are unknown (Bates et al. 2014)
  - asymptotically, test statistics of single coefficients are normally distributed, but with finite sample sizes they do not follow (clear) t-distributions
  - differences in Deviance ( $= -2 \cdot \log(\text{Likelihood})$ ) do not follow (clear) F-distributions
- The lmerTest package provides p-value for fixed effects based on a t-Test (Kuznetsova, Brockhoff, and Christensen 2017)
  - by default it uses Satterthwaite's degrees of freedom method (Satterthwaite 1946)
  - it also can use Kenward and Roger's method for REML estimates (Kenward and Roger 1997)

## Alternative Tests for Single Coefficients

- The t-Test provided by the lmerTest package is one way to test single coefficients from the fixed part of your model
- An alternative is to use LR-Tests
  - LR-Tests compare two models, by comparing the difference in their Likelihood values against a  $\chi^2$ -Distribution, with the DF equal to the number of parameters by which the models differ
  - if we compare two models that differ only by one variable in their fixed part, we get a test of the single coefficient
  - But remember: LR-Tests can only be used on models that differ in their fixed part, if the models are estimates with FML
  - LR-test only work with nested models; this is, the simpler model can be derived from the more complex one by parameter restrictions



## R Example: Testing A Single Coefficient with LR-Tests

```
# A two level-model with two variables
model1 <- lmer(formula=ais ~ 1 + agea + res_mig + (1|Country),
               REML=FALSE, data=essround8)
# A two level-model with three variables
model2 <- lmer(formula=ais ~ 1 + agea + res_mig + gndr + (1|Country),
               REML=FALSE, data=essround8)
# Comparing both models with a LR-Test
anova(model1, model2, test="Chisq")
```

```
## Data: essround8
```

```
## Models:
```

```
## model1: ais ~ 1 + agea + res_mig + (1 | Country)
```

```
## model2: ais ~ 1 + agea + res_mig + gndr + (1 | Country)
```

```
##      npar   AIC   BIC logLik deviance Chisq Df Pr(>Chisq)
```

```
## model1    5 26033 26069 -13012    26023
```

```
## model2    6 26035 26078 -13012    26023 0.013  1    0.9092
```

# Testing Groups of Fixed Coefficients with LR-Tests

- The LR-Test can also be used to compare two models that differ by more than one term in the fixed part of the model
  - again, this requires to use FML instead of REML to estimate the models
  - the test will inform us about the joint significance of the variables that included
- When do we need such tests?
  - with categorical variables that enter the model as a set of dummies, we can test the joint significance of the dummies, i.e. the categorical variable
  - we may have sets of variables that are theoretically grouped (e.g. measures of globalization vs. control variables). We can use LR-Tests to test for the joint significance of a set of variables that corresponds to a theoretically meaningful construct (e.g. are the two variables measuring globalization jointly significant?)

## R Example: Testing Multiple Fixed Coefficients with LR-Tests

Testing the joint significance of the three dummies that measure educational levels:

```
# A two level-model with three variables
model2 <- lmer(formula=ais ~ 1 + agea + res_mig + gnдр + (1|Country),
               REML=FALSE, data=essround8)

# A two level-model with six variables
model3 <- lmer(formula=ais ~ 1 + agea + res_mig + gnдр + res_educ + (1|Country),
               REML=FALSE, data=essround8)

# Comparing both models with a LR-Test
anova(model2, model3, test="Chisq")
```

```
## Data: essround8
## Models:
## model2: ais ~ 1 + agea + res_mig + gnдр + (1 | Country)
## model3: ais ~ 1 + agea + res_mig + gnдр + res_educ + (1 | Country)
##      npar   AIC   BIC logLik deviance  Chisq Df Pr(>Chisq)
## model2    6 26035 26078 -13012    26023
## model3    9 25471 25535 -12726    25453 570.62  3 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

# Testing Variances in Multilevel Models

- Multilevel modeling is all about variances
- Testing variance components, aka the random part of a model, comes with additional difficulties
  - variances are by definition  $\geq 0$ , they cannot be  $< 0$
  - then, the null hypothesis ( $H_0$ ) is on the boundary of the parameter space
  - statistical tests are not valid on the border of the parameter space
  - however, the tests will be conservative. So, if you find a significant effect, it will be also significant in 'reality'
- Some people recommend to divide the p-value for tests of variance components by 2 to get an approximation of the correct value Pinheiro and Bates (2006)

# Testing Variances with LR-Tests

- We can compare two models that differ in their random part with the LR-Test
  - if the two models differ only in the random part, we can use REML to estimate the models
  - since variance components are biased with FML, we should use REML when we test variance components
- Comparing an empty model with a simple OLS model will tell us if it is necessary to use multilevel models at all
  - this is much better than the 10% rule of thumb
- In the next steps, we can test whether additional random effects (random slopes) are necessary, i.e. whether they significantly improve the model fit
  - note that the inclusion of a random slope into your model will increase the number of parameters by more than just one, as we introduce also covariances
  - anova uses the correct DF for the  $\chi^2$ -distribution

# R Example: Testing for Variance Components with LR-Tests

```
# An empty model without REs
model0_ols <- lm(formula=ais ~ 1, data=essround8)
# Adding a random intercept at the country-level
model0 <- lmer(formula=ais ~ 1 + (1|Country), data=essround8)
# Testing whether multilevel models are needed
anova(model0, model0_ols, test="Chisq")

## refitting model(s) with ML (instead of REML)

## Data: essround8
## Models:
## model0_ols: ais ~ 1
## model0: ais ~ 1 + (1 | Country)
##           npar    AIC    BIC logLik deviance Chisq Df Pr(>Chisq)
## model0_ols     2 27524 27538 -13760    27520
## model0         3 26157 26178 -13075    26151  1369  1 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Note: if you want to compare a linear OLS model with a mixed model and use the function `anova` for the test, you have to name the model estimated by `lmer` first and the model estimated by `lm` second (if you don't, you will get an error)

## R Example: Testing the Significance of Random Slopes

```
# A two level-model without random slopes
model4 <- lmer(formula=ais ~ 1 + agea + res_mig + gndr + lrscale +
               (1|Country), data=essround8)

# A two level-model with random slope for left-right
model5 <- lmer(formula=ais ~ 1 + agea + res_mig + gndr + lrscale +
               (1 + lrscale|Country), data=essround8)

# Comparing both models with a LR-Test
anova(model4, model5, test="Chisq", refit=FALSE)

## Data: essround8
## Models:
## model4: ais ~ 1 + agea + res_mig + gndr + lrscale + (1 | Country)
## model5: ais ~ 1 + agea + res_mig + gndr + lrscale + (1 + lrscale | Country)
##      npar   AIC   BIC logLik deviance  Chisq Df Pr(>Chisq)
## model4    7 25798 25848 -12892   25784
## model5    9 25545 25609 -12763   25527 257.05  2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Note: the anova function will always refit your models with FML because it does not 'recognize' when two models differ only in their random part and therefore can be compared with REML estimates. To force this comparison you have to set the argument refit to FALSE

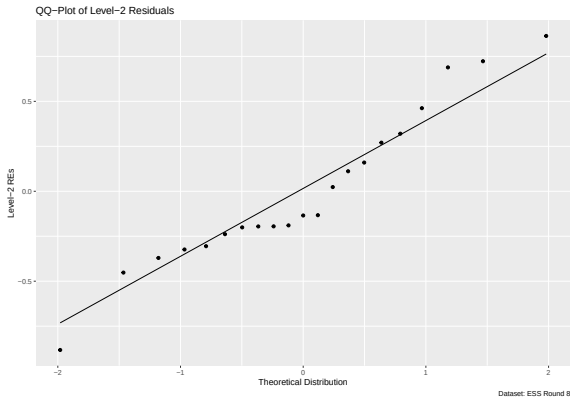
- Checking the distributional assumptions can be particularly important in multilevel models
  - when the number of higher-level clusters is small, the law of large numbers does not work in our favor and the residuals at higher-levels may not be normally distributed
  - this would result in biased standard errors
- With few level-2 units, single outliers (e.g. countries) can have very strong effects on the results
  - graphical inspection can help here to detect outliers
  - an alternative are outlier statistics such as Cook's D or DFBETAs



- You may want to check
  - Schmidt-Catran, Fairbrother, and Andreß (2019) for a general overview
  - Van der Meer, Te Grotenhuis, and Pelzer (2010) discuss the outlier problem in detail and provide an R package ([influence.me](https://influence.me)) for outlier detection in multilevel models
  - Snijders and Berkhof (2008) treat various diagnostics for multilevel models

# R Example: Checking the Normality Assumption

```
# Estimate empty model
model0 <- lmer(formula=ais ~ 1 + lrscale + (1|Country), data=essround8)
u0j <- ranef(model0)
u_df <- as.data.frame(u0j)
ggplot(u_df, aes(sample=condval)) + stat_qq() + stat_qq_line() +
  labs(title="QQ-Plot of Level-2 Residuals", caption="Dataset: ESS Round 8",
       x="Theoretical Distribution", y="Level-2 RES")
```



# No Exercise Today / Thank you

We will not do an exercise today!

- Thanks for your attention!
  - I had fun
  - I hope you learned something
  - (and maybe had some fun as well)

# Literature

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